

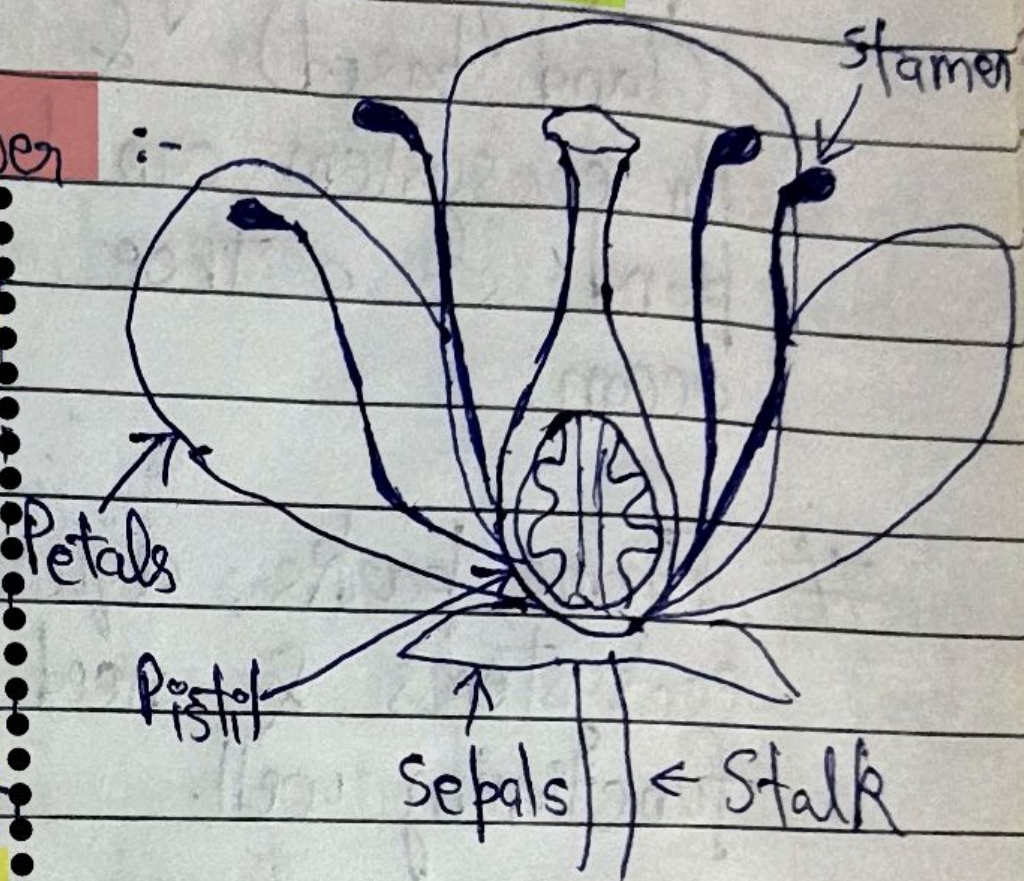
Ch-12

- # Changes in natural habitats, temperature & rainfall patterns are troublesome to animal. It also makes them hostile.
- # A habitat is a place where an organism lives. Plants & animals interact with each other & adapt to their surroundings to survive.
- # The non-living elements of a habitat are known as Abiotic components & living beings of a habitat are known as Biotic components.
- # An organism needs appropriate conditions to survive. A ~~habitat~~ habitat can be inhabited by more than one animal also.
- # The number of an particular organism in a habitat is known as its population.
- # Different living organisms living in the same habitat coexist with harmony.

A community is formed when different populations ~~sharing~~ share the same habitat. All organisms in an community are interlinked & interdependent.

Anatomy of a flower :-

A flower has a stalk, petal (brightly coloured parts for attracting insects), sepal (for protection of flower when it is a bud),



stamen (male reproductive part) & pistils (also known as Carpels; female reproductive part)

Agents like wind, water & insects can carry pollens from stamen to carpels of same (or different) flower. This is known as pollination & it is important for formation of fruits & seeds.

When biotic components of an habitat interact with the abiotic components of that habitat, an ecosystem is ~~for~~ formed. There are mainly 2 types of ecosystems: Terrestrial (land based) & Aquatic (water based).

An ecosystem can be as small as a pond or a tree or as large as an ocean.

Farms, parks, aquariums are human-made ecosystems & need continuous care for functioning well.

Organisms → Autotrophs

↓
Heterotrophs

↙
Herbivores

↘
Omnivores

↓
Carnivores

Food chain is a simple sequence showing 'who eats whom' in an ecosystem.

⇒ Grass → Grasshopper → Frog → Snake → Eagle

Each organism in a food chain has a specific position, called trophic levels.
→ Plants → Bottom-most or first level

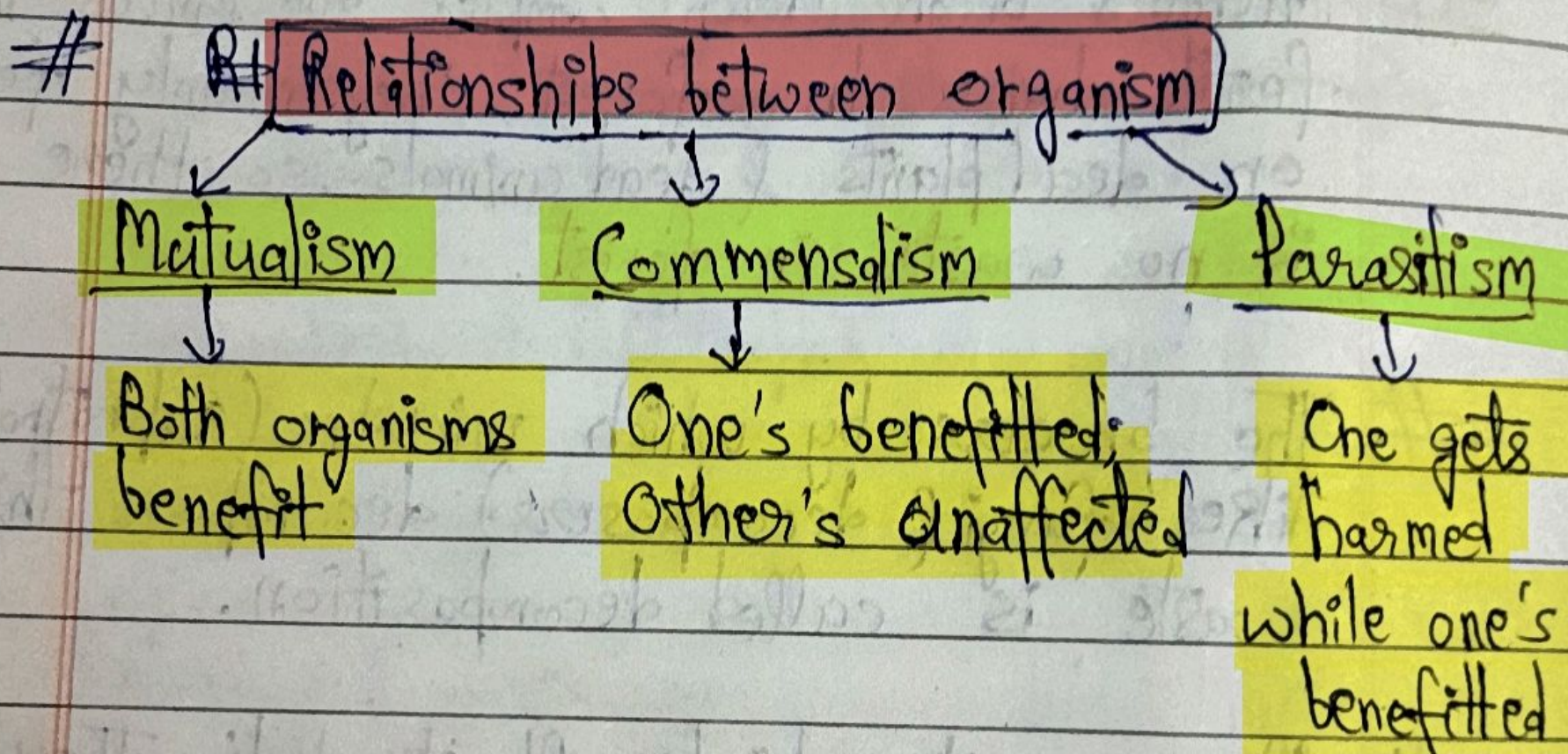
Each of the organisms may be eaten by 2 or more organisms, thus food chains are interlinked with each other, called a food web.

Microbes break down complex substances into fertile humus. In a forest, they mainly feed on dead plants & dead animals, so there is no waste in forest.

The process by which microbes (saprotrophs like fungi & decomposers) decompose this waste is called decomposition.

Many migratory birds fly to India. They migrate to avoid harsh climate. These birds enhance aesthetics & act as pollinators. They are also predators of insects which feed on healthy crops & reduce their growth. Eg: Demoiselle Crane.

Small changes can disturb ecological balance by huge margins. Like - Fewer plants means less food for herbivores & less food for herbivores means less food for carnivore. Eg - Harvesting Indian Bullfrog disturbed ecological balance & led to the increase in the number of increase. Thus, ecological balance can be disturbed by natural or human-made changes.



Ecosystems provide us fresh air, soil, food, water, fibres, timber, medicines, aesthetics & recreational value. Eg - There are many wildlife sanctuaries, national parks, etc., to protect nature. Steps that are being taken to protect Sundarban delta in India

& its mangrove forests, from pollution

B/w 1950 - 1965, India faced a crisis due to low crop production. But, then came the Green Revolution & saved India as in that revolution, farmers started using machine, synthetic fertilisers & pesticides which increased production. But, large-scale using of chemicals is unsustainable as they destroy nutrition of crop & fertility of soil. Growing same type of crop ~~of~~ on the same land repeatedly also reduces soil fertility. Chemicals used in farming also kill microbes that help crop growth & reduce Humus, which makes ~~the~~ the soil prone to erosion.

To tackle all these problems, many farmers are looking towards other options like organic farming. Eg - Sikkim became India's first completely organic state.

Vrikshayurveda is an ancient Indian text about methods of farming & acted like a guide for farming with harmony with nature.